Riemann Sums and Definite Integrals

ANSWER KEY

Approximating area with Riemann sums

1 Estimate $\int_0^4 x^2\,dx$ using a right-endpoint Riemann sum with $n = 4$ subintervals.

$\Delta x = 1$; right endpoints $1, 2, 3, 4$: sum $= 1(1) + 1(4) + 1(9) + 1(16) = 30$.

2 Estimate $\int_1^5 (2x + 1)\,dx$ using a left-endpoint Riemann sum with $n = 4$ subintervals.

$\Delta x = 1$; left endpoints $1, 2, 3, 4$: values $3, 5, 7, 9$: sum $= 24$.

The definite integral and its properties

3 Given $\int_0^3 f(x)\,dx = 8$ and $\int_3^7 f(x)\,dx = -2$, find:

a $\int_0^7 f(x)\,dx \;= 6$

b $\int_7^3 f(x)\,dx \;= 2$

c $\int_0^3 5f(x)\,dx \;= 40$

4 The graph of $f(x) = 4 - x^2$ is shown, together with the region between the curve and the x-axis on $[0, 2]$. Evaluate $\int_0^2 (4 - x^2)\,dx$ exactly.

$\left[4x - \frac{x^3}{3}\right]_0^2 = 8 - \frac{8}{3} = \frac{16}{3}$.

5 Evaluate $\int_{-2}^2 (x^3 - 4x)\,dx$ by recognizing the integrand's symmetry, without computing an antiderivative.

The integrand is odd ($f(-x) = -f(x)$), and the interval is symmetric about 0, so the integral is 0.

Midpoint and trapezoidal approximations

6 Estimate $\int_0^4 x^2\,dx$ using a midpoint Riemann sum with $n = 4$ subintervals, and compare to the exact value $\frac{64}{3} \approx 21.33$.

$\Delta x = 1$; midpoints $0.5, 1.5, 2.5, 3.5$: sum $= 0.25 + 2.25 + 6.25 + 12.25 = 21$. Close to the exact value 21.33.

7 Estimate $\int_1^5 \frac{1}{x}\,dx$ using the trapezoidal rule with $n = 4$ subintervals, to three decimal places.

$\Delta x = 1$, values at $1, 2, 3, 4, 5$: $1, 0.5, 0.333, 0.25, 0.2$. Trapezoidal:
$\frac{1}{2}(1 + 2(0.5 + 0.333 + 0.25) + 0.2) = \frac{1}{2}(1.2 + 2.166) \approx 1.683$.

Signed area

8 The graph of $f(x) = x - 2$ crosses the x-axis at $x = 2$. Evaluate $\int_0^4 (x - 2)\,dx$ and explain why the result does not equal the total area between the curve and the axis.

$\left[\frac{x^2}{2} - 2x\right]_0^4 = (8 - 8) - 0 = 0$. The integral gives signed (net) area: the region below the axis on $[0, 2]$ cancels the region above it on $[2, 4]$, even though both have positive area.

9 Find the total (unsigned) area between $f(x) = x - 2$ and the x-axis on $[0, 4]$, splitting the integral at the sign change $x = 2$.
$\left|\int_0^2 (x-2)\,dx\right| + \left|\int_2^4 (x-2)\,dx\right|$. $\int_0^2 (x-2)\,dx = \left[\frac{x^2}{2} - 2x\right]_0^2 = (2-4) - 0 = -2$, so $|-2| = 2$.
$\int_2^4 (x-2)\,dx = \left[\frac{x^2}{2} - 2x\right]_2^4 = (8-8) - (2-4) = 0 - (-2) = 2$. Total area $= 2 + 2 = 4$.

Comparing Riemann sum estimates

10 For the increasing function $f(x) = x^2$ on $[0, 4]$, will the left-endpoint sum overestimate or underestimate the true area? Justify your answer.

Underestimate — since f is increasing, the left endpoint of each subinterval gives the smallest function value on that subinterval, so each rectangle sits below the curve.

11 Using the right-endpoint result (30) and left-endpoint result $(24$, computed earlier for a different function) is not directly comparable — instead, compare the right-endpoint estimate (30) for $\int_0^4 x^2 dx$ to the exact value $\frac{64}{3} \approx 21.33$, and explain the direction of the error.

The right-endpoint estimate (30) overestimates the true value (≈ 21.33), consistent with $f(x) = x^2$ being increasing — right endpoints give the largest function value on each subinterval, so the rectangles sit above the curve.

Riemann sums from a table of values

12 A table gives $f(0) = 2$, $f(1) = 5$, $f(2) = 9$, $f(3) = 14$, $f(4) = 20$. Use a right-endpoint Riemann sum with these 4 equal subintervals of width 1 to approximate $\int_0^4 f(x)\,dx$.

$R_4 = 1[f(1) + f(2) + f(3) + f(4)] = 5 + 9 + 14 + 20 = 48$.

The definite integral as a limit of Riemann sums

13 Write the formal limit definition of the definite integral $\int_a^b f(x)\,dx$ in terms of a Riemann sum, and explain the role of $n \to \infty$.
$\int_a^b f(x)\,dx = \lim_{n\to\infty} \sum_{i=1}^{n} f(x_i^*)\Delta x$, where $\Delta x = \frac{b-a}{n}$. As $n \to \infty$, the rectangles become infinitely thin, and the sum converges to the exact signed area under the curve, regardless of which sample point x_i^* (left, right, midpoint) is used within each subinterval.

Riemann sums with sigma notation

14 Write the right-endpoint Riemann sum R_n for $\int_1^3 x^2\,dx$ using sigma notation, with $\Delta x = \frac{2}{n}$ and $x_i = 1 + i\Delta x$. Do not evaluate the sum.
$R_n = \sum_{i=1}^{n} f(x_i)\Delta x = \sum_{i=1}^{n} \left(1 + \frac{2i}{n}\right)^2 \cdot \frac{2}{n}$.

15 Estimate $\int_0^6 f(x)\,dx$ using a midpoint Riemann sum with $n = 3$ subintervals, given $f(1) = 4$, $f(3) = 10$, $f(5) = 16$.

$\Delta x = 2$; midpoints are exactly $1, 3, 5$: sum $= 2(4 + 10 + 16) = 60$.

Comparing approximation methods on a concave function

16 For a concave-down function on $[a, b]$, which is larger: the trapezoidal estimate or the midpoint estimate? Justify briefly, and state which one overestimates the true area.

For concave-down f, the trapezoids lie below the curve (underestimate), while midpoint rectangles' tops cross above the curve on average (overestimate). So the midpoint estimate is larger, and the trapezoidal estimate underestimates the true area.